1 **(a)** $0.15 + 0.25 + x + 0.2 + 3x = 1$ **[M1]**

$x = 0.1$ **[A1]**

(b) $P(> 3) = P(4) + P(5) = 0.2 + 3(0.1) = 0.5$ **[M1]**

Expected number $= 250 \times 0.5 = 125$ **[A1]**

[Total Marks: 4]

2 **(a)** $1 - (0.15 + 0.3 + 0.25 + 0.1)$ or $1 - 0.8$ **[M1]**

0.2 **[A1]**

(b) 200×0.3 **[M1]**

60 **[A1]**

[Total Marks: 4]

3 **(a)** $\frac{70}{200}$ or $\frac{7}{20}$ or 0.35 **[B1]**

(b) $1000 \times \frac{45}{200}$ **[M1]**

225 **[A1]**

(c) No, because the relative frequencies are not all close to 0.25 (or 50 out of 200) **[B1]**

(d) A larger number of trials leads to a relative frequency that is closer to the theoretical probability **[B1]**

[Total Marks: 5]

4 **(a)** 0.27 **[B1]**

(b) If the dice were fair, each score would be expected to occur 40 times (or with relative frequency 0.20), whereas the observed frequencies differ considerably from 40 **[B1]**

(c) Total frequency of odd scores $= 32 + 54 + 40 = 126$ **[M1]**

Relative frequency of an odd score $= \frac{126}{200} = 0.63$ **[M1]**

$0.63 \times 500 = 315$ **[A1]**

(d) $\frac{40}{200} \times n = 36$ (or $0.2n = 36$) **[M1]**

180 **[A1]**

[Total Marks: 7]

5 **(a)** $x + 2x + 5 + 3x - 2 + x + 8 = 60$ **[M1]**

$7x + 11 = 60 \implies 7x = 49$

$x = 7$ **[A1]**

(b) Frequency of $3 = 3(7) - 2 = 19$

$\frac{19}{60}$ **[B1]**

(c) Frequency of $2 = 2(7) + 5 = 19$

Relative frequency of $2 = \frac{19}{60}$

$1200 \times \frac{19}{60}$ **[M1]**

380 **[A1]**

[Total Marks: 5]

6 (a) $\frac{2x+4}{x^2+10x} = \frac{1}{6}$ giving $x^2 + 10x = 6(2x + 4)$ **[M1]**

$x^2 + 10x = 12x + 24$ simplifying to $x^2 - 2x - 24 = 0$ **[A1]**

(b) $(x - 6)(x + 4) = 0$ **[M1]**

$x = 6$ (rejecting $x = -4$ since $x > 0$) **[A1]**

Total spins $= 6^2 + 10(6) = 96$ **[A1]**

(c) (i) Sector B occurs $5(6) + 2 = 32$ times, giving $\frac{32}{96}$ **[M1]**

$\frac{1}{3}$ **[A1]**

(ii) Sector C occurs $96 - 16 - 32 = 48$ times, giving $\frac{48}{96}$ **[M1]**

$\frac{1}{2}$ **[A1]**

(d) Expected number of spins: $A = 600 \times \frac{1}{6} = 100$, $B = 600 \times \frac{1}{3} = 200$, $C = 600 \times \frac{1}{2} = 300$ **[M1]**

Expected net profit $= (100 \times 6) + (200 \times 2) - (300 \times 3) = 600 + 400 - 900$ **[M1]**

100 **[A1]**

[Total Marks: 12]

7 (a) $x = 1 - (0.15 + 0.45 + 0.10)$

$x = 0.30$ **[B1]**

(b) 300×0.45 **[M1]**

135 **[A1]**

(c) Biased, because the relative frequencies are not all close to 0.25 (or specifically Blue is 0.45, which is much higher than 0.25) **[B1]**

[Total Marks: 4]

8 (a) 75 **[B1]**

(b) $\frac{30}{75}$ or $\frac{30}{30+45}$ **[M1]**

0.4 or $\frac{2}{5}$ **[A1]**

(c) $P(\text{tail}) = 1 - 0.4 = 0.6$ or $\frac{45}{75}$ **[B1]**

300×0.6 **[M1]**

180 **[A1]**

[Total Marks: 6]

9 (a) $\frac{x}{300} = 0.22 \implies x = 300 \times 0.22$ **[M1]**

$y = 300 - (84 + 105 + 66)$ **[M1]**

$x = 66$, $y = 45$ **[A1]**

(b) Relative frequency of Yellow $= \frac{45}{300}$ or 0.15 **[M1]**

Total beads $= \frac{15}{0.15} = 100$ **[A1]**

(c) Probability of Blue or Green $= \frac{105+66}{300} = \frac{171}{300}$ or 0.57 **[M1]**

$800 \times 0.57 = 456$ **[A1]**

[Total Marks: 7]

10 (a) $\frac{260}{800}$ **[M1]**

$\frac{13}{40}$ **[A1]**

(b) $260 + 180 + w + w + 40 = 800 \implies 480 + 2w = 800$ **[M1]**

$w = 160$ **[A1]**

(c) $\frac{160}{800}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(d) $P(\text{Black}) = \frac{180}{800} = \frac{9}{40}$ **[M1]**

Expected number $= 12000 \times \frac{9}{40} = 2700$ **[A1]**

[Total Marks: 7]

11 (a) $\frac{y+35}{250} = 0.42$ giving $y + 35 = 105$ **[M1]**

$y = 70$ **[A1]**

$x = 250 - (15 + 95 + 70 + 35) = 35$ **[A1]**

(b) Number of light bulbs with $t \leq 1000$ in sample $= 15 + 35 = 50$ **[M1]**

Estimated number $= \frac{50}{250} \times 4800$ **[M1]**

960 **[A1]**

(c) To avoid bias so that the sample is representative of all light bulbs produced by the factory **[B1]**

(d) $\frac{35+115}{250+750} = \frac{150}{1000}$ **[M1]**

0.15 (or $\frac{3}{20}$) **[A1]**

[Total Marks: 9]

12 (a) $0.025 \times N = 40$ or $N = \frac{40}{0.025}$ **[M1]**

1600 **[A1]**

(b) Expected defective $= 2400 - 2364 = 36$ **[B1]**

New probability of defective $= \frac{36}{2400} = 0.015$ **[B1]**

Decrease in probability $= 0.025 - 0.015 = 0.01$ **[M1]**

Percentage decrease $= \frac{0.01}{0.025} \times 100$ **[M1]**

40 **[A1]**

[Total Marks: 7]

13 (a) $0.15 + 2x + x + 0.05 + 0.2 + x = 1$ **[M1]**

$4x + 0.4 = 1 \implies 4x = 0.6 \implies x = 0.15$ **[A1]**

(b) $P(2, 3, 5) = 2(0.15) + (0.15 + 0.05) + 0.15 = 0.3 + 0.2 + 0.15 = 0.65$ **[M1]**

$600 \times 0.65 = 390$ **[A1]**

(c) Sets up both equations: $N \times p = 54$ and $N \times (1 - 3p) = 36$ **[M1]**

Expands the second equation: $N - 3Np = 36$ **[M1]**

Substitutes $Np = 54$ into the expanded equation: $N - 3(54) = 36$ **[M1]**

$N = 198$, $p = \frac{54}{198} = \frac{3}{11}$ (or 0.272...) **[A1]**

[Total Marks: 8]

14 (a) Sum of probabilities: $3x + 2y = 1 \implies y = \frac{1 - 3x}{2}$ **[B1]**

$E_1 = 400x$ and $E_3 = 400y \implies 400x \times 400y = 5600 \implies xy = \frac{7}{200}$ **[M1]**

Substitute for y: $x\left(\frac{1-3x}{2}\right) = \frac{7}{200} \implies 100x(1 - 3x) = 7 \implies 300x^2 - 100x + 7 = 0$

[A1]

(b) Factorising $(30x - 7)(10x - 1) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = \frac{7}{30}$ and $x = \frac{1}{10}$ (or 0.1) **[A1]**

When $x = \frac{7}{30}$, $y = 0.15$ (reject as $y < x$). When $x = 0.1$, $y = 0.35$ (accept as $y > x$) **[M1]**

$x = 0.1, y = 0.35$ **[A1]**

(c) $P(\text{even}) = 2x + y = 2(0.1) + 0.35 = 0.55$ **[M1]**

Expected number $= 400 \times 0.55 = 220$ **[A1]**

[Total Marks: 9]

15 **(a)** $300p + 200(p + 0.15) = 255$ **[M1]**

$300p + 200p + 30 = 255$ **[M1]**

$500p = 225 \implies p = \frac{225}{500}$

$p = 0.45$ **[A1]**

(b) Probability Coin B is Heads $= 0.45 + 0.15 = 0.60$

Probability of different outcomes = Probability (H, T) + Probability (T, H)

$(0.45 \times 0.40) + (0.55 \times 0.60)$ **[M1]**

$0.18 + 0.33 = 0.51$ **[M1]**

$1000 \times 0.51 = 510$ **[A1]**

(c) **(i)** $\frac{72}{N} = 0.48$ **[M1]**

$N = \frac{72}{0.48}$

$N = 150$ **[A1]**

(ii) Probability of Tails $= 1 - 0.45 = 0.55$

150×0.55 **[M1]**

82.5 **[A1]**

[Total Marks: 10]